Math GU4053 Intro to Algebraic Topology
Homework 4

due: Wednesday, February 25, 2026, before 23:59pm

submit the completed assignment to https://www.gradescope.com

---

- This assignment has **3 questions** on **two pages**.
- This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
- Your solutions must be *handwritten*. For instance on a tablet/iPad, or with pen and paper scanned to a pdf.
- While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
- Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
- Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)

---

1. Let $n \geq 2$. In class, we have seen *real projective space* $\mathbb{R}P^n = S^n/\sim$, where $\sim$ is the equivalence relation generated by $x \sim -x$ for all $x \in S^n$. We also saw that its fundamental group is isomorphic to $\mathbb{Z}/2\mathbb{Z}$, which may be used without proof in this problem.

Prove that any continuous map

$$f : \mathbb{R}P^n \to S^1$$

can be written as $f(x) = (\cos(2\pi\tilde{f}(x)), \sin(2\pi\tilde{f}(x)))$ for some continuous $\tilde{f} : \mathbb{R}P^n \to \mathbb{R}$.

2. In class, we defined the Klein bottle as $K = \mathbb{R}^2/G$, for a certain group

$$G = \{T^a S^b \mid (a,b) \in \mathbb{Z}^2\}.$$

We also defined (see [GW§2.6.3]) an action of this group on $\mathbb{R}^2$ and proved that it is a covering space action.

(a) Prove that the function $\phi : G \to \mathbb{Z}/2\mathbb{Z}$, defined as $\phi(T^a S^b) = a \mod 2$, is a homomorphism, and deduce that $H = \{T^{2a} S^b \mid \mathbb{Z}^2\}$ is a subgroup of G.

(b) Let T denote the torus. Deduce the existence of a covering map

$$p : T \to K,$$

such that the subgroup

$$\mathrm{Im}(p_* : \pi_1(T, x) \to \pi_1(K, p(x)))$$

has index 2, for a suitable base point $x \in T$.

Hint: a problem on HW3 may be useful.

3. In a previous course on algebra, you have seen the classification of subgroups $H < \mathbb{Z}$: these are of the form $H = d\mathbb{Z}$ for a uniquely determined $d \in \mathbb{Z}_{\geq 0}$. For instance, $d = 0$ corresponds to the trivial subgroup, $d = 1$ corresponds to the entire group, $d = 2$ corresponds to the subgroup consisting of even numbers, etc.

In this problem we discuss some aspects of the classification question for subgroups of the additive group $\mathbb{Z}^2$. Let $H < \mathbb{Z}^2$ be such a subgroup. Let $\pi : \mathbb{Z}^2 \to \mathbb{Z}$ and $j : \mathbb{Z} \to \mathbb{Z}^2$ be given by $\pi(x, y) = x$ and $j(y) = (0, y)$. Then we have

$$\pi(H) = d\mathbb{Z}$$
$$j^{-1}(H) = e\mathbb{Z}$$

for uniquely determined $d, e \in \mathbb{Z}_{\geq 0}$.

(a) Assume $d > 0$ and $e > 0$. Prove that there exists an $n \in \{0, \ldots, e-1\}$ such that H is generated by the two elements (d, n) and $(0, e)$.

(b) Assume $d > 0$ and $e = 0$. Prove that H is infinite cyclic and that H is generated by $(d, n) \in \mathbb{Z}^2$ for some $n \in \mathbb{Z}$.

(c) Assume $d = 0$ and $e > 0$. Prove that H is infinite cyclic and that H is generated by $(0, e) \in \mathbb{Z}^2$.

Hint for (i): one way to begin is to first argue that there exists $n \geq 0$ such that $(d, n) \in H$, choose the smallest such n, and then show $n < e$ and that (d, n) and $(0, e)$ generate all of H.

Remark (not needed for doing the problem): to complete the "classification" of subgroups of $\mathbb{Z}^2$, one should include the remaining case case $d = e = 0$ and also discuss uniqueness. For instance, in case (a) there is precisely one n with the stated properties.